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import cv2
import numpy as np

# Create a white canvas
canvas = np.ones((700, 900, 3), dtype=np.uint8) * 255

def draw_shapes(choice):
    global canvas

    if choice == 1:
        cv2.line(canvas, (100, 100), (400, 100), (0, 0, 255), 5)
        print("Line Drawn")

    elif choice == 2:
        cv2.rectangle(canvas, (100, 150), (350, 300), (0, 255, 0), 4)
        print("Rectangle Drawn")

    elif choice == 3:
        cv2.circle(canvas, (550, 220), 80, (0, 0, 255), 5)
        print("Circle Drawn")

    elif choice == 4:
        cv2.ellipse(canvas, (550, 450), (120, 60), 30, 0, 360, (255, 0, 255), 4)
        print("Ellipse Drawn")

    elif choice == 5:
        pts = np.array([[150,450],[250,350],[350,450],[300,600],[200,600]], np.int32)
        pts = pts.reshape((-1,1,2))
        cv2.polylines(canvas, [pts], True, (0,165,255), 4)
        print("Polygon Drawn")

    elif choice == 6:
        cv2.arrowedLine(canvas, (450,600), (800,600), (128,0,128), 5)
        print("Arrowed Line Drawn")

    elif choice == 7:
        cv2.putText(canvas,
                    "OpenCV Drawing App",
                    (180,50),
                    cv2.FONT_HERSHEY_COMPLEX,
                    1.2,
                    (0,0,0),
                    3)
        print("Text Added")

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elif choice == 8:  
    cv2.imwrite("OpenCV_Drawing_Output.png", canvas)  
    print("Image Saved Successfully!")
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elif choice == 9:  
    canvas = np.ones((700,900,3), dtype=np.uint8) * 255  
    print("Canvas Cleared")
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else:  
    print("Invalid Choice")
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while True:
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    print("\n===== OpenCV Drawing App =====")  
    print("1. Draw Line")  
    print("2. Draw Rectangle")  
    print("3. Draw Circle")  
    print("4. Draw Ellipse")  
    print("5. Draw Polygon")  
    print("6. Draw Arrowed Line")  
    print("7. Add Text")  
    print("8. Save Image")  
    print("9. Clear Canvas")  
    print("0. Exit")
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    choice = int(input("Enter your choice: "))
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    if choice == 0:  
        break
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    draw_shapes(choice)
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    cv2.imshow("OpenCV Drawing Canvas", canvas)
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    print("Press any key to continue...")  
    cv2.waitKey(0)
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cv2.destroyAllWindows()
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